Noel Andolz Aguado

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Portfolio Website

EDUCATION

Carlos III University of Madrid

Telecommunications engineering & cybersecurity dual master 2025–2027

Bachelor's thesis - "Measurement and Characterization of Radio Signals in Amplitude and Phase" 2024–2025

Telecommunication technologies enigneering at San Francisco State University 2022 – 2023

Academic year 2022-2023 at San Francisco State University in the framework of Universidad Carlos III de Madrid Non European Movility program.

Bachelor's Degree in Telecommunication Technologies Engineering 2020 – 2025

Professional Experience

Internship — Ericsson **April 2025 – May 2026**

- Contributing to the testing team for Ericsson's Cloud Native 5G solutions, with a focus on enhancing the efficiency of the system testing phase through data modeling and artificial intelligence techniques.

Internship — Accenture **October 2024 – February 2025**

- As an Intern in the Tech Advance Testing team, I contributed to creating and implementing comprehensive testing strategies.

Area VIP Real Madrid **September 2023 – August 2025**

- As a VIP Area Attendant at Santiago Bernabéu Stadium, I am responsible for ensuring that our distinguished guests experience the full extent of our premium services.

Waiter and Barman — La Ciboulette **June 2020 – July 2024**

- As a student, gaining practical experience has been a top priority for me. That's why I decided to work at Ciboulette Catering, where I've learned about teamwork, serving procedures, and event organization. Working alongside a dedicated team has taught me the importance of collaboration, and how it leads to better outcomes for everyone involved.

PROJECTS

Mininet and Ryu **2025 (view project)**

- Development of Software-Defined Networking (SDN) scenarios using Mininet and the Ryu controller. The project involved the emulation of network topologies, configuration of OpenFlow-based switches, and implementation of custom control logic in Python, including MAC learning, flow table management, and packet forwarding strategies.

  Additionally, the work covered automation of network experiments through the Mininet Python API, integration with remote controllers, and analysis of protocol behavior (e.g., ARP and OpenFlow) in controlled virtual environments.

SDN Development Lab **2025 (view project)**

- Design and implementation of an IPv4 software router using Ryu within a Mininet-based emulated environment. The project involved developing custom OpenFlow control logic in Python to perform Layer 3 packet forwarding, including routing table matching, TTL manipulation, and dynamic header rewriting (MAC/IP fields).

  The implementation was extended to support control-plane functionalities such as ICMP echo reply generation and dynamic ARP resolution, enabling full end-to-end connectivity without static address configuration. The system was validated through traffic analysis and packet inspection, ensuring correct protocol behavior and compliance with real router operations.

IP Route Lookup using Compressed Trie **2023 (view project)**

- An IP route lookup implementation using the compressed trie (Patricia trie) algorithm. This program efficiently searches for the longest prefix match in a routing table for given IP addresses.

Participation on Vodafone Challenge **2023**

- As a part of my Mobile Communications class, I have participated in the 2023 Vodafone Challenge. We had the great opportunity to visit Vodafone Building in Madrid where we were shown Vodafone's current operations and their strategic vision for the near future as well as being lectured about 5G, Massive MIMO, RAN and Open RAN,... Finally, we were given a group challenge regarding Low Layer Split options in Downlink in which we had to work for the following months. Once the due date had arrived, I had the chance to present alongside my fellow colleague Jose María Ruíz Crespo in front of José Eugenio Caballero. Overall, it has been an enriching experience where teamwork in a real-world environment was key and in which my knowledge about the topic grew immensely.

Dijkstra Algorithm **2022 (view project)**

- A software project developed for the ENGR 476 Computer Communications Networks course. This program implements the Dijkstra's Algorithm to compute shortest paths, using a cost matrix as input where non-connected nodes are represented with a finite cost value (500) instead of infinity.

STM32 Reaction Games **2021 (view project)**

- The firmware implements two two-player reaction games controlled through external buttons, LEDs, a buzzer, and a potentiometer, all connected to the NUCLEO-L152RE development board. Players interact with the hardware peripherals in real time; the microcontroller measures reaction times, drives visual and audible feedback, and outputs results over USART to a serial console.

  The project is structured around four incremental milestones that progressively integrate GPIO configuration, hardware timer interrupts (TIM2/TIM4), ADC-based potentiometer reading, PWM-driven buzzer melodies, and UART serial communication.

Process Scheduler Emulator **2021 (view project)**

- A C-based application that emulates the simplified behavior of a process scheduler, allowing users to interactively manage a scheduling table. This project was developed as part of the Systems Architecture course at Carlos III University of Madrid.

ADDITIONAL INFORMATION

International Exchange:

- San Francisco, California, USA 2022
- Maryland, USA 2019
- Totnes, England 2017

Other studies:

- Music studies at Escuela - Conservatorio Manuel Rodriguez Sales.
- Videogames programming course hosted by Comunidad de Madrid.
- Robotics course hosted by IES ISAAC ALBENIZ.

IT Skills:

C, Java, Python, MATLAB, VHDL, GitHub, ANSYS HFSS, SQL, LTSpice, Excel, AWR

Volunteering:

- Red Cross PINEO and OTL programs 2019 – 2020
- Participation in Scouts groups 101 and 601

LANGUAGES

English: Advanced professional capabilities
Spanish: Native
German: Basic